

BCE-C612
SEMESTER EXAMINATION-MAY2023
CLASS: B.TECH SEMESTER: VI
FORMAL LANGUAGES AND AUTOMATA THEORY

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A

(SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: Define Non-Deterministic Finite Automata

Question-2: Define context free grammar

Question-3: Convert regular expression (01^*+1) to finite automata

Question-4: What is Chomsky Normal Form? Discuss.

Question-5: Bring out the differences of Melay and Moore Machine?

Question-6: Define Push Down Automata.

Question-7: Explain about types of Turing Machine.

Question-8: Discuss in detail about P and NP problems

Question-9: What is undecidable problem?

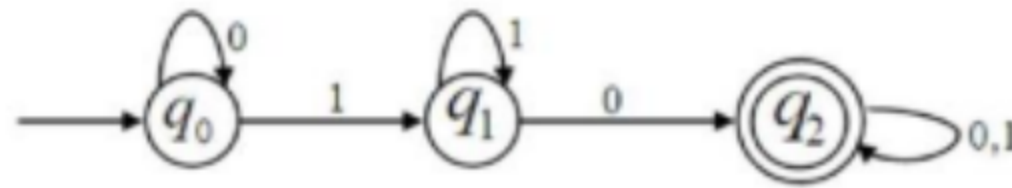
Question-10: Distinguish between class P and class NP Problems.

SECTION-B
(LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 X 10 = 40 Marks)

- Question-1: Design a DFA which accepts set of all strings which are divisible by 5 for binary number.
- Question-2: Explain about Pumping Lemma and write the Applications of the Pumping Lemma?
- Question-3: Define Regular Expression? Explain about the Properties of Regular Expressions.
- Question-4: Construct NFA with ϵ which accepts a language consisting the strings of any number of 0's followed by any number of 1's followed by any number of 2's And also convert into NFA without ϵ transitions.
- Question-5: Construct the regular expression corresponding to the language accepted by following DFA.



- Question-6: For the Grammar $\{S \rightarrow AS/a, A \rightarrow SbA/SS/ba\}$ construct Left most derivation for the string aabbaaa?
- Question-7: Define PDA and construct PDA from the following CFG
 $S \rightarrow aAA$
 $A \rightarrow aS|bS|a$
- Question-8: Define Turing Machine and Construct a Turing Machine that will accept the Language consists of all palindromes of 0's and 1's?